

Choose any 6 questions from the 8 questions to answer (indicate the question number before answering)

Please write your name and student number at the top of your answer sheet !

1. List the intensity transformation functions you know.
2. Give a single intensity transformation function for spreading the intensities of an image so the lowest intensity is 0 and the highest is $L-1$.
3. Briefly introduce Histogram equalization and how to implement it.
4. Explain why the discrete histogram equalization technique does not, in general, yield a flat histogram.
5. Please introduce the sharpening spatial filter and the smoothing spatial filter.
6. Give two examples of weighted average smoothing filter.
7. Write an expression for 2-D discrete convolution.
8. Give examples of Laplacian operator, Sobel operator, and Robert operator respectively.